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WS 2026/2027

Game Theory

Problem Set 2

Deadline Nov 16, 2026, 09:00, OLAT upload*

Problem 1: Two firms compete in quantities

Suppose that there exists a duopoly on the market for smart glasses. The market price depends on the total quantity produced by the two firms, given by $p = 10 - q_1 - q_2$. Both firms can choose a small, medium, or large quantity, $q_i \in \{1, 3, 5\}$. The constant marginal costs are $c_1 = 2$ and $c_2 = 3$. Each firm's profit is $\pi_i = (p - c_i)q_i$. The firms choose their quantities simultaneously.

a. Describe the situation as an extensive-form game and in the strategic form.

b. Apply the iterated elimination of strictly dominated strategies (IEDS). Which strategies survive?

Problem 2: Beliefs and Best Responses

Consider the following static game. The first payoff in each cell belongs to player 1 and the second to player 2.

	L	C	R
U	$(3, 2)$	$(0, 4)$	$(6, 1)$
M	$(2, 5)$	$(5, 1)$	$(1, 3)$
D	$(4, 0)$	$(2, 3)$	$(3, 6)$

Please provide not only the results but also your calculations.

a. What are $u_1(U, R)$, $u_2(M, L)$ and $u_1(D, C)$?

b. Suppose player 1 believes player 2 chooses (L, C, R) with probabilities $\theta_2 = (1/2, 1/4, 1/4)$. Calculate the expected payoff of each pure strategy of player 1 and find $BR_1(\theta_2)$.

c. Suppose player 2 believes player 1 chooses (U, M, D) with probabilities $\theta_1 = (1/2, 0, 1/2)$. Find $BR_2(\theta_1)$.

d. Calculate $u_2(\sigma_1, \sigma_2)$ for $\sigma_1 = (1/2, 0, 1/2)$ and $\sigma_2 = (0, 1/2, 1/2)$.

*Naming scheme: *gt_ws26_ps2_matrikel1_matrikel2_matrikel3.pdf* (use your 7-digit matriculation number, **not** your "s-number" or library ID).

Problem 3: Dominated Strategies & Nash Equilibria .....

Consider the following static 4×4 game. Player 1 chooses a row, and player 2 chooses a column. In each cell, the first payoff belongs to player 1.

	a	b	c	d
A	$(4,4)$	$(0,0)$	$(5,1)$	$(0,0)$
B	$(0,0)$	$(4,4)$	$(5,1)$	$(0,0)$
C	$(0,0)$	$(0,0)$	$(2,1)$	$(0,0)$
D	$(1,0)$	$(1,0)$	$(5,3)$	$(0,0)$

Remember to consider domination by mixed strategies as well.

a. Which strategies survive the iterated elimination of strictly dominated strategies (IEDS)?

b. Which strategies survive the iterated elimination of weakly dominated strategies (IEWDS)? If a mixed strategy is used, give its mixing probabilities.

c. Find the pure Nash equilibria of the game. Compare them with your answers above.

Problem 4: Joining a student initiative .....

Three students decide at the same time whether to join a student initiative (J) or stay home (H). Staying home gives a payoff of 0. Joining gives student i a payoff of

$$u_i(J, s_{-i}) = b_i + 2k_i - 4,$$

where k_i is the number of other participants. Let $(b_1, b_2, b_3) = (5, 3, 1)$.

a. Apply IEDS. Which strategies survive? Show the steps of elimination.

b. Now let $b_1 = 3$, keeping $b_2 = 3$ and $b_3 = 1$. Find all pure Nash equilibria.